Hosting Capacity and the Data-Center Connection Game

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An unprecedented surge in demand for electricity

- **Scale.** ERCOT is tracking more than 438 GW of connection requests from large loads as of mid-2026, nearly 90% of them data centers, against an all-time peak demand of 85.5 GW set on August 2023.[1]
- **Single sites are now GW-sized.**
  - OpenAI's Stargate: nearly 7 GW planned across its announced sites.[2]
  - **ANTHROP\C**: $50B of new data centers in Texas and New York;[3] up to 5 GW of compute with Amazon;[4] single training sites of 2 GW (2027) and 5 GW (2028) expected.[5]

[1] [1] ERCOT, *Trending Topics: ERCOT's New Batch Connection Process for Large Electricity Users*, 2026.

[2] [2] OpenAI, *OpenAI, Oracle, and SoftBank Expand Stargate with Five New AI Data Center Sites*, 2025.

[3] [3] Anthropic, *Anthropic Invests $50 Billion in American AI Infrastructure*, 2025.

[4] [4] Anthropic, *Anthropic and Amazon Expand Collaboration for up to 5 Gigawatts of New Compute*, 2026.

[5] [5] Anthropic, *Build AI in America: Anthropic Energy Report*, 2025.

Who decides, and the two questions for today

- In ERCOT, the transmission service provider (e.g., Oncor) and ERCOT study each large-load request.
- Since 2026, ERCOT studies eligible large loads *together* and allocates transmission capacity "based on what the grid can reliably support".[6]

1. How much load can a bus take? — Part I: hosting capacity
2. When requests compete, who gets served? — Part II: the connection game

[6] [1] ERCOT, *Trending Topics: ERCOT's New Batch Connection Process for Large Electricity Users*, 2026.

Part I: Hosting capacity

What is hosting capacity?

Hosting capacity of a location

The largest amount of new *load* that can be connected there without violating the network's operating limits, with the existing network and controls.

- The idea started with *distributed generation*, e.g. rooftop solar photovoltaic (PV) panels on distribution feeders: how much PV can a feeder absorb?[7]
- The same question applies to large new consumption,[8] and to transmission-level loads such as data centers.[9]
- Computed by stochastic simulation, streamlined rules, time series, or optimization (today).[10]

[7] [6] Bollen et al., *Integration of Distributed Generation in the Power System*, 2011, [7] Rylander et al., EPRI, 2015.

[8] [8] Bollen et al., *Energies*, 2017.

[9] [9] Cicilio et al., *IEEE Transactions on Sustainable Energy*, 2021, [10] Wan et al., arXiv:2610.18405, 2026.

[10] [11] Mulenga et al., *International Journal of Electrical Power & Energy Systems*, 2020, [12] Ismael et al., *Renewable Energy*, 2019.

Single-bus hosting capacity as a linear program

Extra load L at bus k in hour t; existing generators redispatch:[11]

$$\mathrm{HC}_{k,t} = \max_{L,\,\mathbf{p}^{\mathrm{g}},\,\mathbf{p},\,\mathbf{f}} \; L$$

$$\text{s.t.} \quad \underbrace{\mathbf{p} = C\,\mathbf{p}^{\mathrm{g}} - \mathbf{d}_t - L\,\mathbf{e}_k}_{\text{net injections (new load at } k)}, \qquad \underbrace{\mathbf{1}^{\top}\mathbf{p} = 0}_{\text{system balance}}, \qquad \underbrace{\mathbf{f} = H\,\mathbf{p}}_{\text{DC power flow}}$$

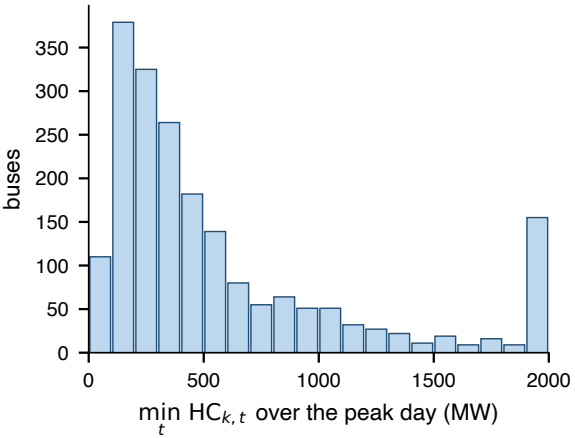
$$\underbrace{|\mathbf{f}| \le \mathbf{F}}_{\text{line thermal limits}}, \qquad \underbrace{\mathbf{p}^{\min} \le \mathbf{p}^{\mathrm{g}} \le \mathbf{p}^{\max}}_{\text{generator limits}},$$

$$L \ge 0$$

Decision variables: L new load, $\mathbf{p}^{\mathrm{g}}$ generator outputs, $\mathbf{p}$ net bus injections, $\mathbf{f}$ line flows. Data: C maps generators to buses, $\mathbf{d}_t$ bus loads, H the PTDF (shift-factor) matrix, $\mathbf{F}$ line limits.

[11] [13] Alturki et al., *Applied Energy*, 2018, [10] Wan et al., arXiv:2610.18405, 2026.

Hosting capacity varies a lot across buses

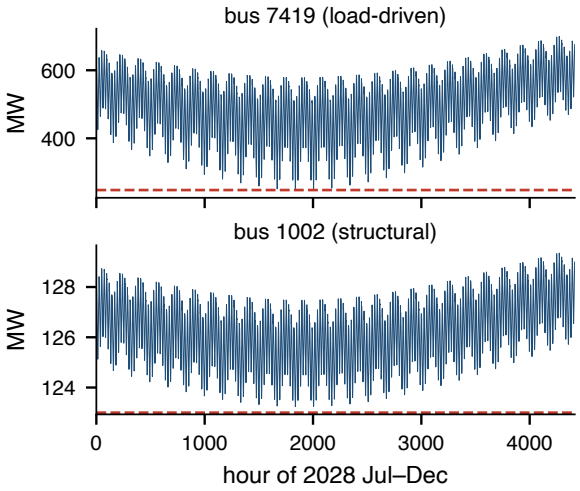


ACTIVSg2000 synthetic Texas grid,[a] 2028 Jul–Dec, the 24 hours of the peak-load day, all 2,000 buses:

- median 372 MW
- 10% of buses below 130 MW
- 90% below 1,536 MW; 146 buses above 2,000 MW (not shown)
- smallest: 10 MW

[a] [14] Birchfield et al., *IEEE Transactions on Power Systems*, 2017.

Over time: the stressed hours bind



A constant load must be served in *every* hour:

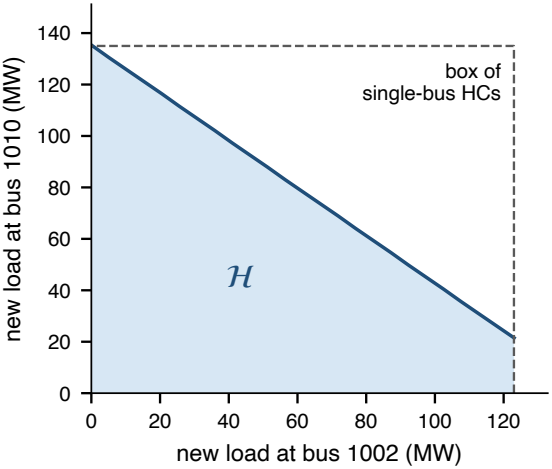
$$\mathrm{HC}_k = \min_t \mathrm{HC}_{k,t} \quad \text{(dashed)}$$

- Bus 7419: 248 MW to 699 MW, median 503 MW: tight at peak hours.
- Bus 1002: 123 MW to 129 MW: the network limits it, not the load cycle.

"Dynamic hosting capacity" treats this time variation explicitly.[a]

[a] [15] Oliveira et al., *Energies*, 2019.

Many buses at once: the hosting-capacity region



$$\mathcal{H} = \left\{ \mathbf{x} \geq 0 : \begin{array}{c} \forall t \ \exists \mathbf{p}^{\mathrm{g}}{}_t, \mathbf{p}_t : \\ \mathbf{p}_t = C\,\mathbf{p}^{\mathrm{g}}{}_t - \mathbf{d}_t - \mathbf{x}, \quad \mathbf{1}^\top \mathbf{p}_t = 0, \\ |H\mathbf{p}_t| \leq \mathbf{F}, \quad \mathbf{p}^{\min} \leq \mathbf{p}^{\mathrm{g}}{}_t \leq \mathbf{p}^{\max} \end{array} \right\}$$

The LP's constraints with new loads $\mathbf{x}$ (one per bus) in place of $L\,\mathbf{e}_k$.

- A convex polytope: the projection of the DCOPF feasible set.[a]
- Single-bus HCs are its axis intercepts; they do not add up.[b]
- Peak hour: bus 1002 hosts 123 MW alone, only 81 MW with 60 MW at neighbor 1010.

[a] [16] Gong et al., *Electric Power Systems Research*, 2024.

[b] [10] Wan et al., arXiv:2610.18405, 2026.

With a local generator

Site at bus k: load L, private generator $0 \le g_t \le G$. The grid only sees the *net* withdrawal $L - g_t$:

$$\text{feasible} \iff \exists\, g_t \in [0, G] : \quad \underbrace{L - g_t}_{\text{net withdrawal}} \le \mathrm{HC}_{k,t} \ \ \forall t \quad \Longrightarrow \quad \mathrm{HC}_k^{G} = \min_t \mathrm{HC}_{k,t} + G.$$

- Exact when the generator is always available; with outages, ramping, or fuel limits, $\mathrm{HC}_k^{G} \le \mathrm{HC}_k + G$.
- The generator runs only when the grid is short or when its cost is below the local price.

With storage: the hours become coupled

Battery at bus k: power P, energy B, signed output u_t (positive = discharge), time step Δt; $\mathrm{HC}_{k,t}$ comes from the single-bus LP:

$$\mathrm{HC}_k^S = \max_{L,\,u,\,e} \; L \quad \text{s.t.} \qquad \underbrace{L - u_t \le \mathrm{HC}_{k,t}}_{\text{grid limit on net withdrawal}}, \qquad \underbrace{-P \le u_t \le P}_{\text{power}},$$

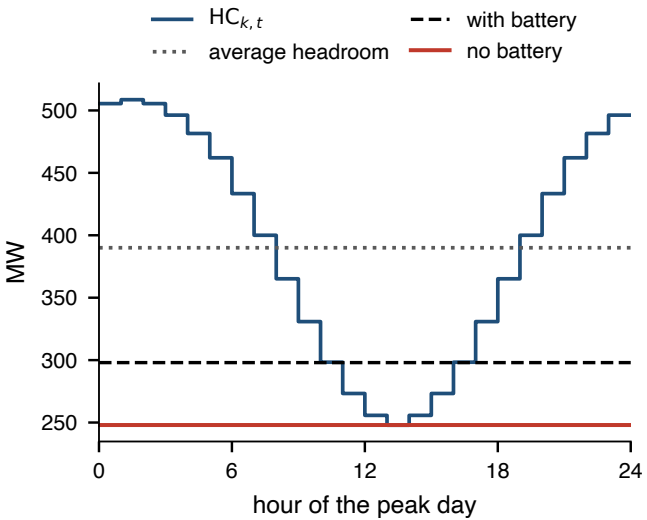
$$\underbrace{e_t = e_{t-1} - \Delta t\, u_t, \quad 0 \le e_t \le B}_{\text{energy}}, \qquad \underbrace{e_0 = e_T}_{\text{daily cycle}}.$$

Decision variables: L new load, u_t battery output, e_t stored energy.

- A multi-period LP: discharge in tight hours, recharge in slack ones.[12]
- Summing over the cycle ($\sum_t u_t = 0$): $\mathrm{HC}_k^S \le \frac{1}{T}\sum_t \mathrm{HC}_{k,t}$, so storage lifts the limit at most from the minimum to the average headroom.
- Power ($L \le \mathrm{HC}_{k,t} + P$) and energy ($B$) keep it below that.

[12] [17] Macedo et al., *IEEE PES Innovative Smart Grid Technologies Conference (ISGT)*, 2017.

Storage over one day

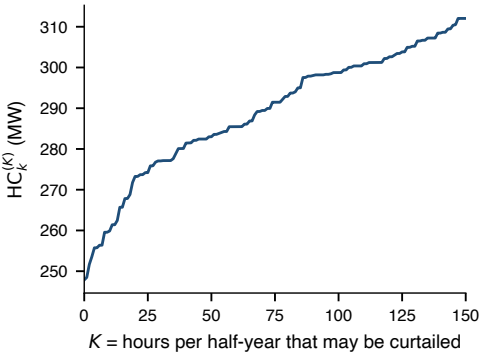


Bus 7419, peak day, hourly steps, daily cycle ($e_0 = e_{24} = B/2$), $B = 4P$:

- no battery: 248 MW
- average headroom (dotted): 390 MW, the upper limit
- $P = 50\,\mathrm{MW}$, $B = 200\,\mathrm{MWh}$: **298** MW

B limits how long a shortfall the battery can cover. Half-year: 248 $\rightarrow$ 298 MW.

Flexibility: curtail in a few hours



Flexible connection:[a]

$$\max_{L,\,\delta} \; L$$

$$\text{s.t. } L - \delta_t \le \mathrm{HC}_{k,t}, \quad \delta \ge 0, \quad \|\delta\|_0 \le K$$

- δ_t is the load curtailed in hour t; nonconvex ($\|\delta\|_0$ counts curtailed hours).
- $\mathrm{HC}_k^{(K)} = \mathrm{HC}_{k,[K+1]}$, the $(K+1)$-th *smallest* hourly value.
- Bus 7419: 248 MW at $K = 0$, 274 MW at $K = 24$ (+11%), 299 MW at $K = 100$ (+21%). Diminishing returns in K.[b]

[a] [18] Gu et al., *Hawaii International Conference on System Sciences (HICSS)*, 2026.

[b] [19] Etherden et al., *IEEE PowerTech*, 2011, [20] Sarker et al., arXiv:2604.20185, 2026.

How much should we trust a hosting-capacity number?

- **Security.** Dropping N-1 contingencies overstates data-center hosting capacity by about 20% on the IEEE RTS-24.[13]
- **Dynamics.** The data center's own ride-through limits (UPS, converters) can be the binding constraint.[14]
- **Model.** DC power flow has no voltage or reactive-power limits and no losses.
- **Simultaneity.** Single-bus numbers ignore everyone else who is connecting at the same time.

The last point leads to the second question: who gets served?

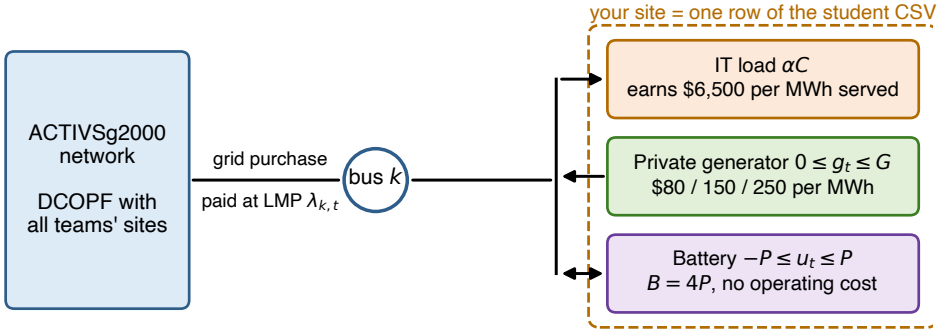
[13] [10] Wan et al., arXiv:2610.18405, 2026.

[14] [21] Ren et al., arXiv:2609.03030, 2026.

Part II: The connection game

The game

- Teams are data-center operators on ACTIVSg2000 (2,000 buses, 3,206 branches).
- One CSV row per site: bus, connection C, generator $G \leq C$, battery P with $B = 4P$.
- Six-hour periods (mean of 6 hours), 4 per day, 724–736 per half-year.
- The instructor solves one joint DCOPF with every team's sites.



Schedule and scoring

Week	Submissions	Counts?
1	2027 Jan–Jun, Jul–Dec	practice
2	2028 Jan–Jun, Jul–Dec	both
3	2029 Jan–Jun, Jul–Dec	both

- Sites built in Submission 1 stay for Submission 2 and are not charged again.
- Submission 1 pays no energy cost.

score = IT value − energy − gen. operating − capital

IT value	\$6,500 per MWh served
energy	LMP × grid purchase
generator operation	\$80 / 150 / 250 per MWh
site	\$25M per MW
generator capital	\$1.5M per MW
storage capital	\$0.4M/MW + \$0.25M/MWh

Break-even share of load served

about 44% over a week if built in Submission 1;
87% if added in Submission 2.

The evaluator

For each day d and period $t = 1, \dots, 4$ ($\Delta t = 6\,\text{h}$): a DCOPF with all teams' sites, coupled through each battery's daily cycle.

$$\min \ \sum_{d,t} \Delta t \left(\mathbf{c}^{\top} \mathbf{p}^{\mathrm{g}}{}_{dt} + \sum_s \sum_m c^m q^m_{s,dt} \right)$$

s.t. DCOPF balance and line limits with loads $\alpha_i C_s$ and private q_s, u_s

$$e_{s,d,t} = e_{s,d,t-1} - \Delta t \, u_{s,dt}, \quad 0 \le e \le B_s, \quad |u| \le P_s, \quad e_{s,d,0} = e_{s,d,4} = B_s/2$$

$$\sum_{s \in i,\, k(s)=n} u_{s,dt} \le \sum_{s \in i,\, k(s)=n} \alpha_i C_s \quad \text{(a team's assets serve only its own load)}$$

- An exact LP that separates by day (Benders or one LP per day).
- LMP = dual of nodal balance; it prices energy in Submission 2.

Who gets served when requests exceed the region?

The game's evaluator uses the *uniform rule*, never first-come-first-served:

1. **Common level:** the largest t such that serving $\min(r_i, t)$ to every team i is feasible in every period (r_i = requested MW, one fraction α_i for all of i's sites).
2. **Water-filling:** teams that are not blocked keep rising until each is fully served or blocked.
3. **Minimum allocation:** a team served $< 50\%$ of its request gets nothing (largest dropped first), as in ERCOT's batch study.

Uniform rule is strategy-proof for one resource;[15] on a network an inflated request can lower everyone's level, hence step 3.

- Capital is charged on what you *asked* for, not on what is served.
- Real counterparts: interconnection queues (first come, first served), ERCOT's 2026 batch allocation of transmission capacity,[16] fairness rules for curtailing distributed generation.[17]

[15] [22] Sprumont, *Econometrica*, 1991.

[16] [1] ERCOT, *Trending Topics: ERCOT's New Batch Connection Process for Large Electricity Users*, 2026.

[17] [23] Liu et al., *IEEE Transactions on Smart Grid*, 2020, [24] Gebbran et al., *Applied Energy*, 2021.

A two-team example on the real grid

Team A at bus 1002, team B at neighbor 1010; each picks 50 or 100 MW, built in Submission 1. Peak hour, every hour treated alike; payoffs per week ($M):

A \ B	B small (50)	B large (100)
A small (50)	1,601 / 1,601	1,601 / 2,580
A large (100)	2,759 / 1,601	1,511 / 1,506

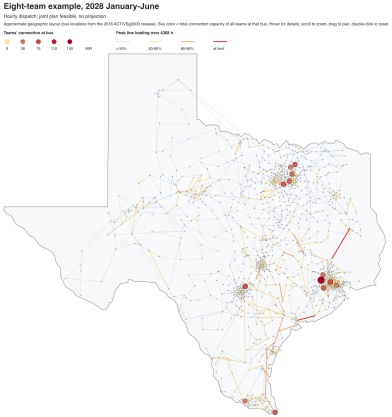
- One large, one small: the small team is served in full, the large one gets the rest (92.2 or 89.1 MW) and pays for 100 MW.
- Both large: both cut to the same level (A 70.4 MW, B 70.3 MW); both lose.
- Nash equilibria: (A small, B large); (A large, B small). A game of *chicken*.

What students get, and a live demo

- `ercot-game-check-csv`: format and bus checks
- `ercot-game-preview`: approximate score, plan alone
- `ercot-game-hosting-capacity`: the LP from this lecture
- After each round: score breakdown, every team's α, projection on/off, and a grid map

```
ercot-game-hosting-capacity \
  --artifact-dir .../2028/jul_dec \
  --start-period 304 --period-count 4 \
  --bus-id 1002 --bus-id 1010 --output-dir
     demo/hc
```

Demo on the students' six-hour package.



Eight-team example, hourly dispatch: bus color = total MW of all teams, line color = peak loading; approximate locations.

Takeaways

- Hosting capacity is an LP: the DCOPF with a "maximize the new load" objective.
- Time: a constant load is limited by the worst hour. Generation adds G; storage moves the limit toward the average; flexibility toward the $(K+1)$-th worst.
- Space: single-bus numbers are intercepts of a region and do not add up.
- With many applicants, who gets served is a design choice (queue, batch allocation, proportional fairness, prices), and it turns planning into a game.

Discussion. How should a grid operator allocate scarce headroom among data centers? What would you change in the game's projection rule?

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[2] OpenAI. *OpenAI, Oracle, and SoftBank Expand Stargate with Five New AI Data Center Sites*. Sept. 2025.

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